



HSE RESEARCH SEMINAR
Data Analysis in Finance

Limits to Arbitrage in Electricity Markets

A case study of MISO

Birge et al. (2017) · replication and a 2024–2026 extension

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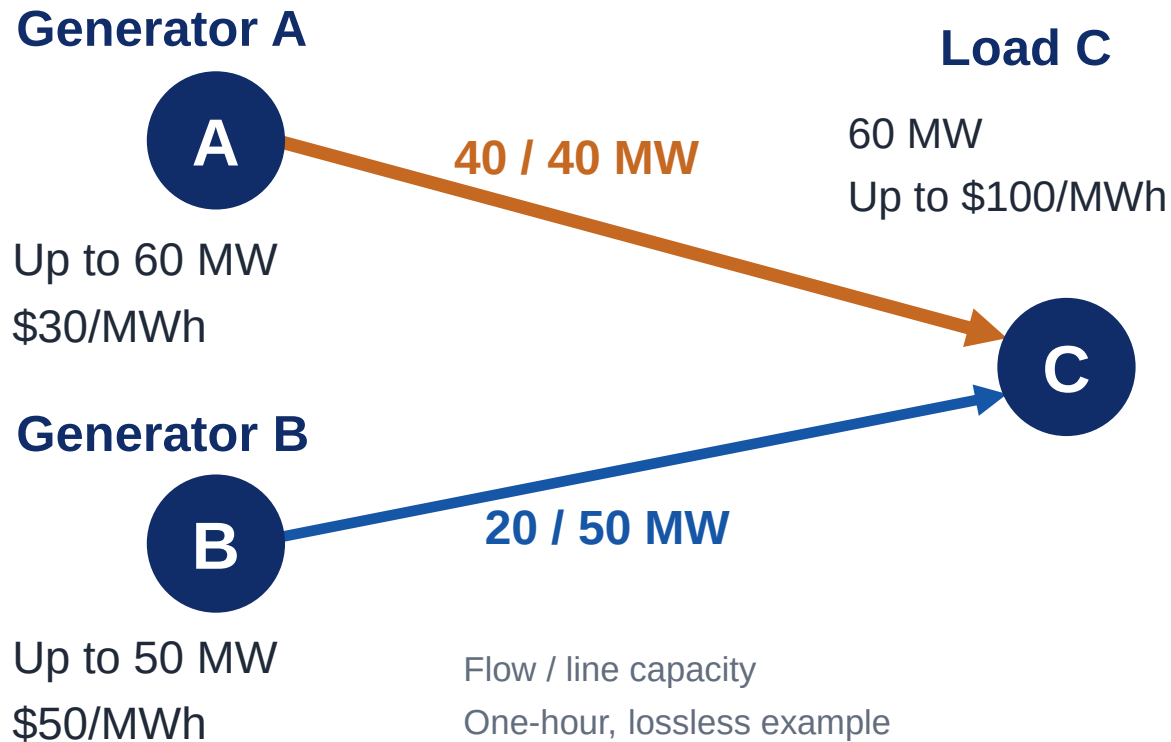
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September 2026



Electricity is traded on a constrained network

MISO clears a central auction while respecting the physics of the grid.



$$G = (V, E)$$

V: buses with generation or load
E: transmission lines

Bids: node, hour, MW and price
Generators also report start-up costs and operating limits.

$$|\text{flow}| \leq \text{line capacity}$$



Day-ahead planning supports real-time delivery

Generators need time to start; buyers want to limit exposure to uncertain prices.

DAY-AHEAD / DA

Today, for each hour tomorrow

Schedule generation early.

Agree a price for scheduled energy.



REAL-TIME / RT

At delivery, every five minutes

Balance actual supply and demand.

Settle deviations at the RT price.

A buyer schedules 60 MWh at \$70, then consumes 65 MWh when RT is \$120:

$$60 \times \$70 + 5 \times \$120 = \$4,800$$



Virtual trades connect DA and RT prices

Financial positions enter the day-ahead auction and settle in the opposite direction at real-time prices.

VIRTUAL SUPPLY

Sell in DA → buy back in RT

$$\text{Profit} = \text{DA} - \text{RT} - \text{charges}$$

Benefits when DA is sufficiently higher

VIRTUAL DEMAND

Buy in DA → sell back in RT

$$\text{Profit} = \text{RT} - \text{DA} - \text{charges}$$

Benefits when RT is sufficiently higher

One MWh, same node and delivery hour. No physical generation or consumption.

Expected $\text{DA} > \text{RT}$ → more virtual supply → pressure for prices to converge



Observation 1: the day-ahead premium persists

Premium = DA price – RT price, in USD/MWh. Positive values favour virtual supply before charges.

2008–2012



2024–2026 · through August

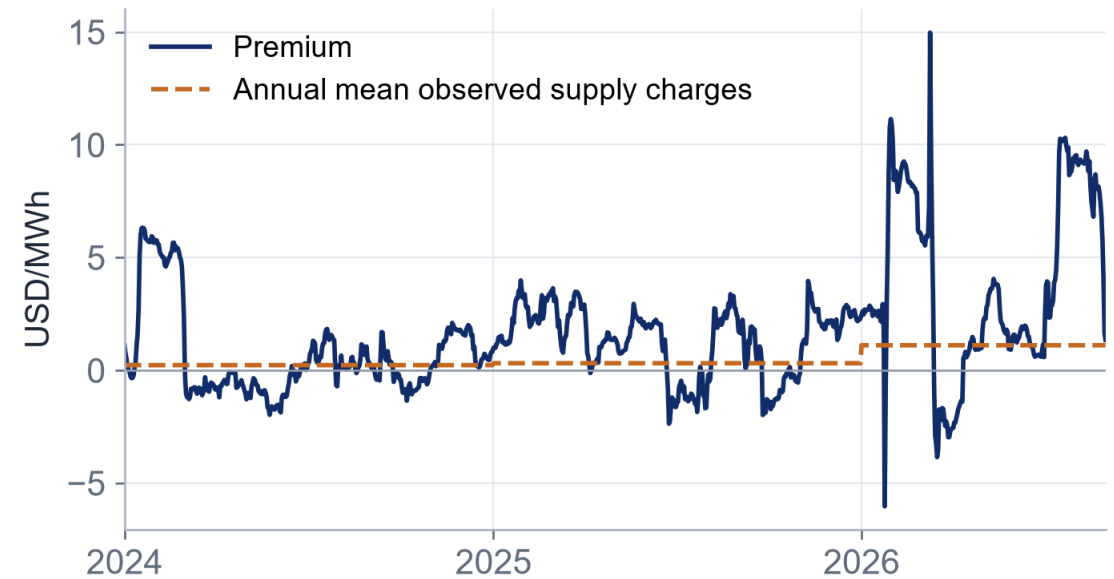
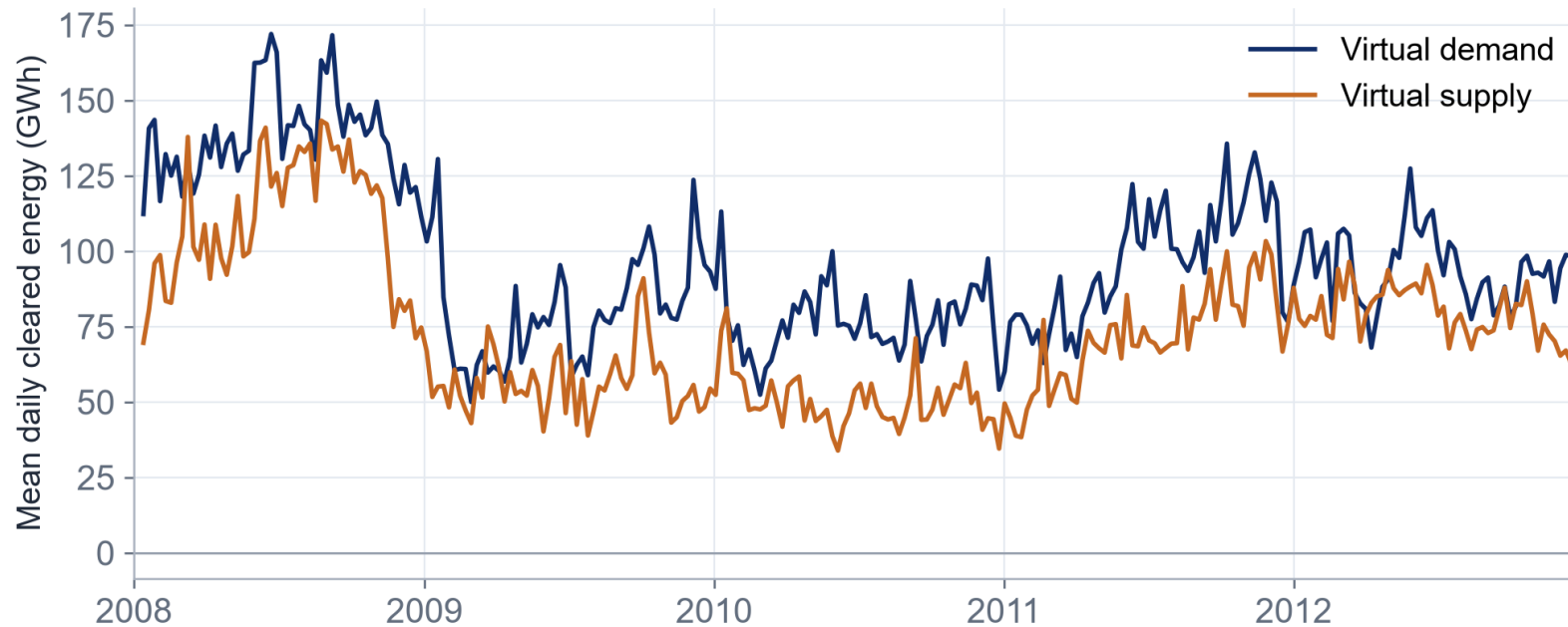


Fig. 1 reconstruction. 45-day mean of daily means of hourly cross-node medians. Different y-axis ranges; recent charges are partial.

Observation 2: virtual demand exceeds supply

Traders collectively buy more day-ahead energy than they sell, despite the positive average premium.



+23.7

GWh per day
more virtual demand

**Who keeps buying
the more expensive
market?**

Fig. 2 reconstruction, 2008–2012. Weekly means of daily cleared volumes. Aggregate volumes include physical participants' virtual bids.



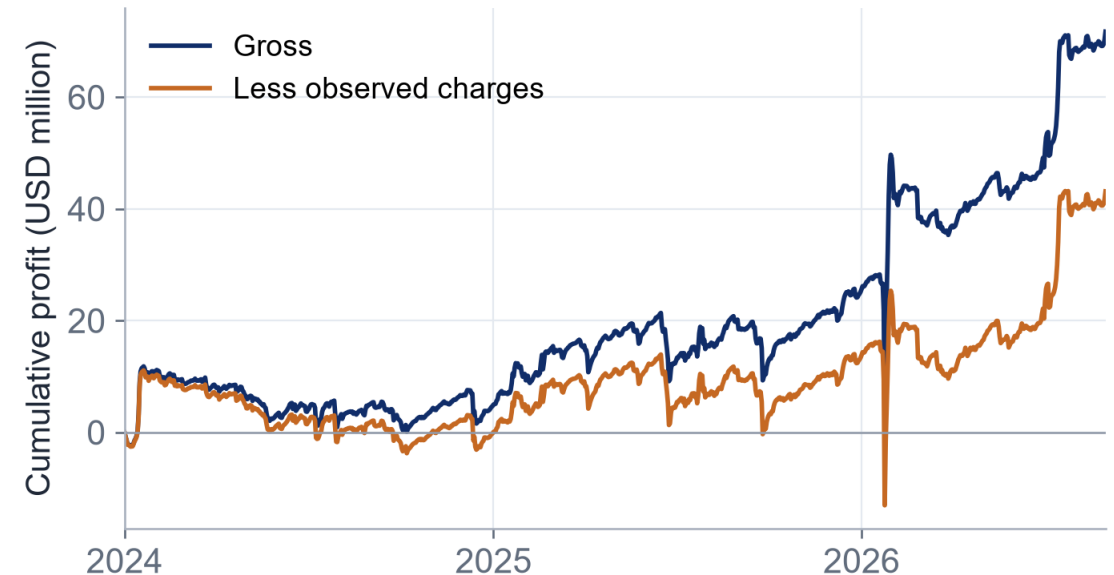
Observation 3: cumulative profits hide large losses

Diagnostic strategy: sell one MWh at every published node, every hour; settle the reverse position in RT.

2008–2012 · gross only



2024–2026 · through August



Recent result: +\$43.1m after observed charges | Worst day: -\$22.6m (24 Jan 2026)

Fig. 3-style extension. Assumes full clearing and no price impact or financing costs. Historical fees unavailable; recent fees partly missing.

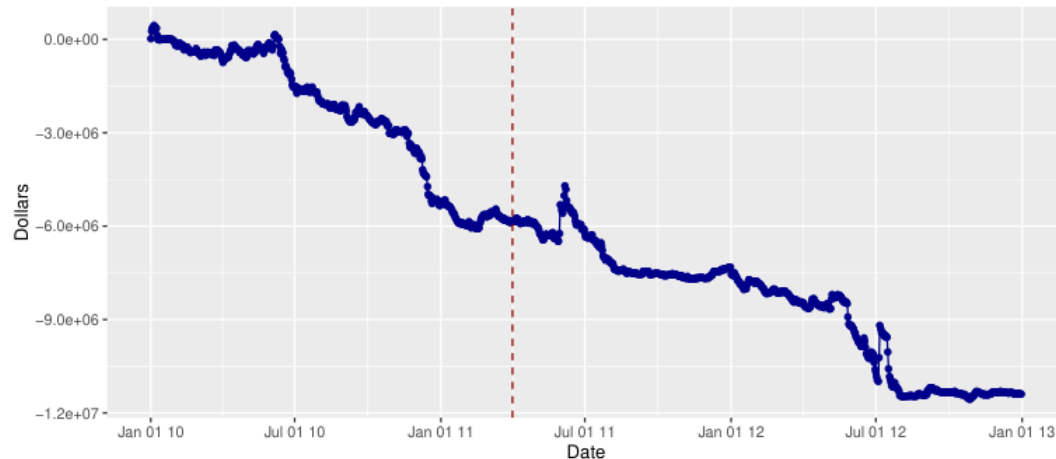
An FTR links profits to congestion between nodes

Financial transmission right: a contract paying the DA congestion difference between its sink and source.

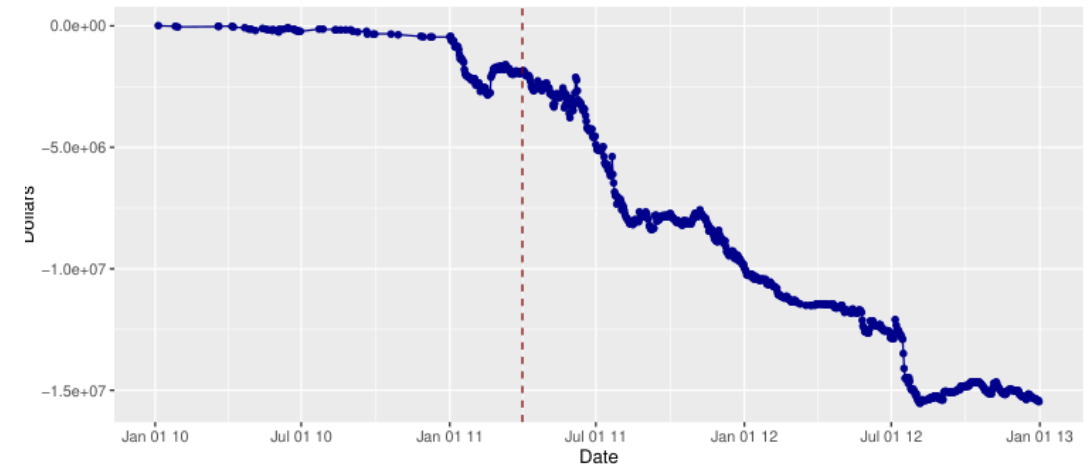


Original paper, Fig. 9: virtual trading losses continue for years

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Source: Birge et al. (2017), Fig. 9 (two original panels, after RSG); MISO FTR description. Other positions may explain virtual losses.



Experiment: add an FTR to a losing virtual position

An accounting counterfactual. It does not establish common ownership or estimate the price effect of a bid.

Modern illustration

10 MW · MICHIGAN.HUB → INDIANA.HUB

Winter 2024–2025 · 1,008 peak hours

FTR entitlement less auction cost **+\$5,535**

Buy 1 MWh virtually at the sink each hour **−\$1,552**

Combined result **+\$3,983**

Entitlement \$16,424.30 – auction cost \$10,889.50

Paper: the same logic

Mean USD per FTR observation

Before April 2011

$\$1,059 - \$818 = \$241$

After April 2011

$\$1,139 - \$917 = \$222$



Conclusions: fees, capital and linked positions matter

The premium is only one part of the trader's economic exposure.

WHAT THE PAPER ARGUES

OUR ASSESSMENT

01 Financial trading can reduce the day-ahead premium.

Historical premium estimates match the paper's annual regression table.

02 Fees and capital constraints limit this arbitrage activity.

Modern gains coexist with severe daily losses; cost coverage remains incomplete.

03 FTR incentives may sustain loss-making virtual positions.

Combined gains are plausible; causality and ownership remain unknown.

Assess a trading opportunity through its full payoff, costs and capital needs.